Interfacing DAC with 8051

LAB # 12

Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

"On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Submitted to:

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Date:

1st June 2024

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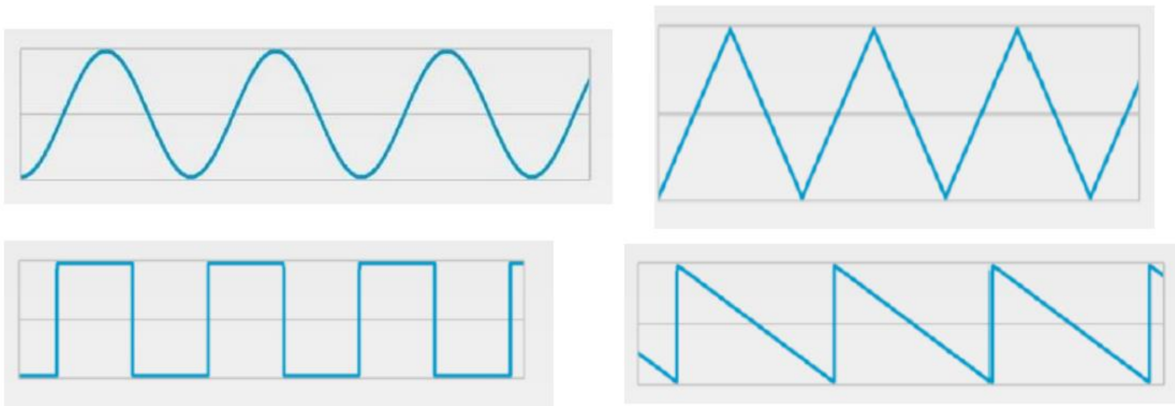
Task 1:

Generate the following waveform



You can use a button, when pressed once then generate triangular wave, when pressed again generate the square wave and so on. Use Port-interrupt for button detection.

Code:

Source Code

main.c

```
#include<reg51.h>

char state;
sbit btn = P3^2;
int i;

void sinewave();
void squarewave();
void sawtoothwave();
void triangularwave();
void delay();

void int_fun() interrupt 0{
        state++;
        if(state > 3 | state < 0 )
                state = 0;
}


void main(void){
                EA = 1;
                EX0 = 1;
                IT0 = 1; //Edge triggered interrupt
                P1 = 0x00;
                P2 =0x00;

                while(1){
```

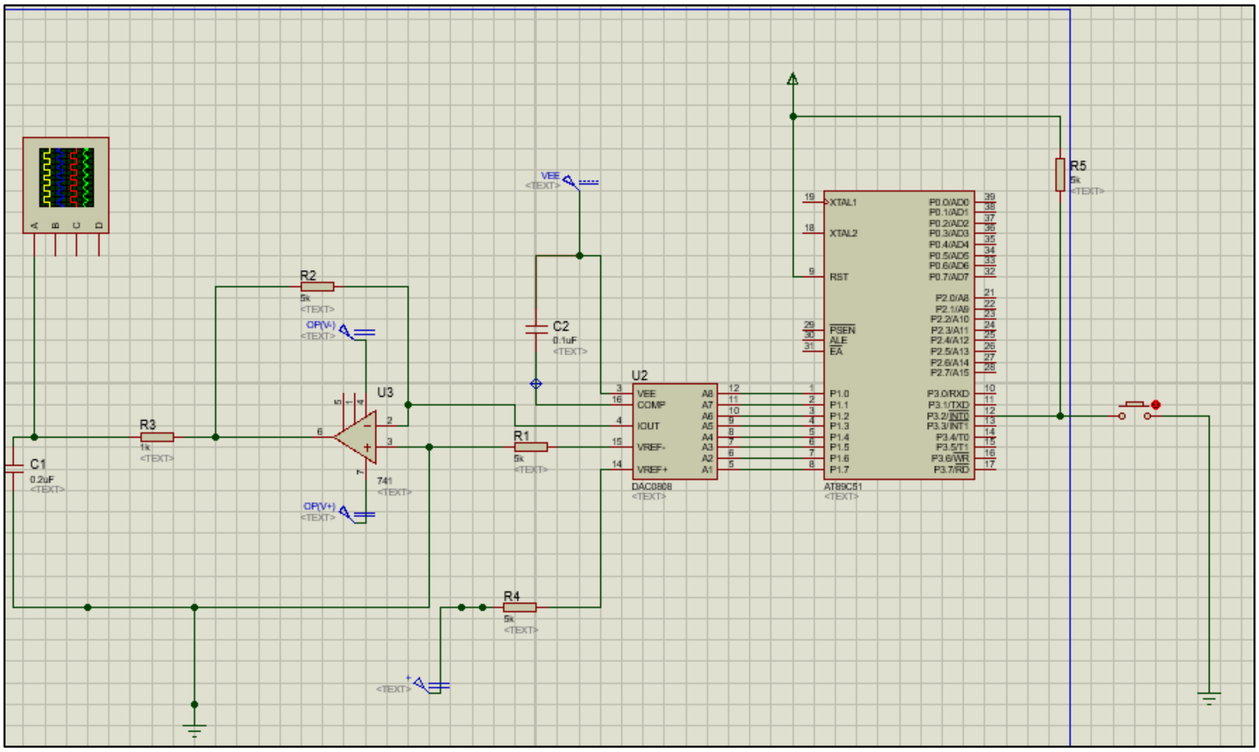
main.c

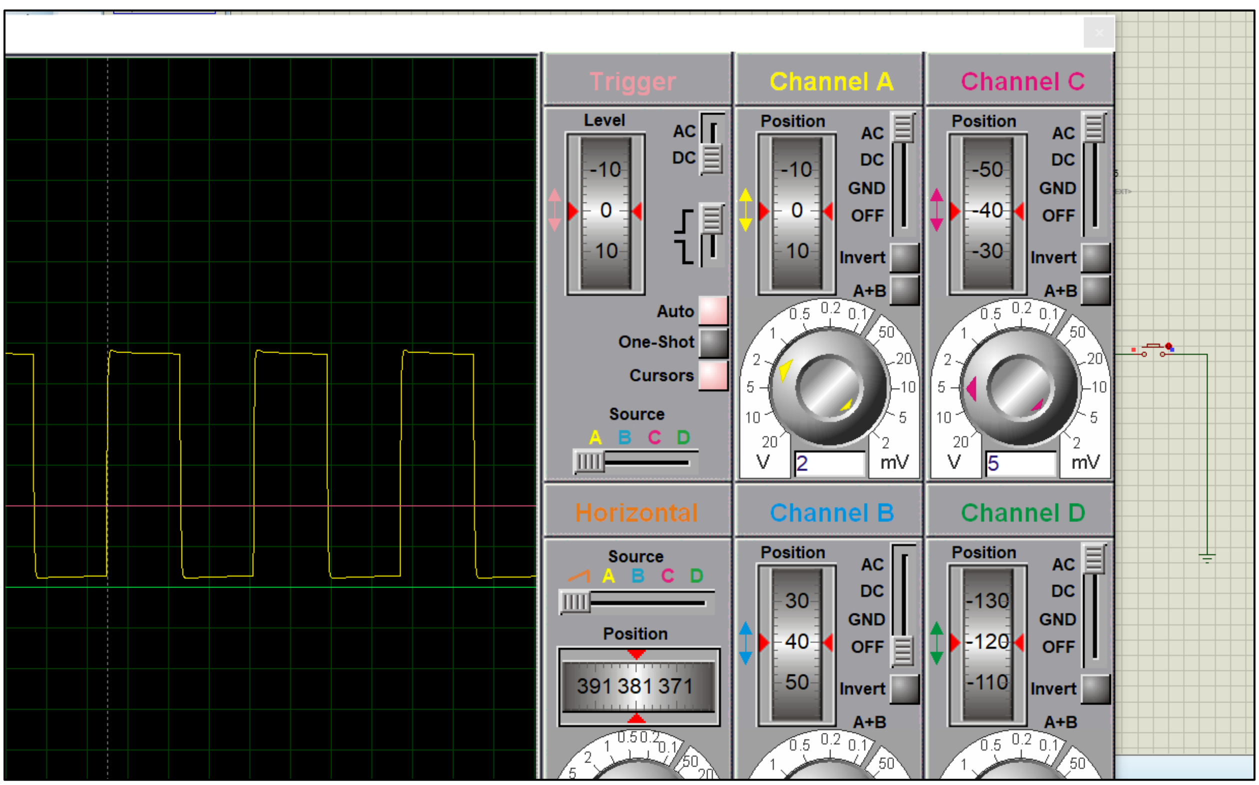
```c
		while(1){
		  P2 =state;
		  if(state == 0)
			squarewave();

		  else if(state == 1)
			sinewave();

		  else if(state == 2)
			sawtoothwave();

		  else if(state == 3)
			triangularwave();
		}
}

void sinewave(){


	char sine_val[21] = {0, 13, 20, 54, 76, 100, 132, 180, 210, 233, 255, 233, 210, 180, 132, 100, 76, 54, 20, 13, 0 };

	for(i = 0; i<21; i++){
		P1 = sine_val[i];
	}
}

void squarewave(){
```

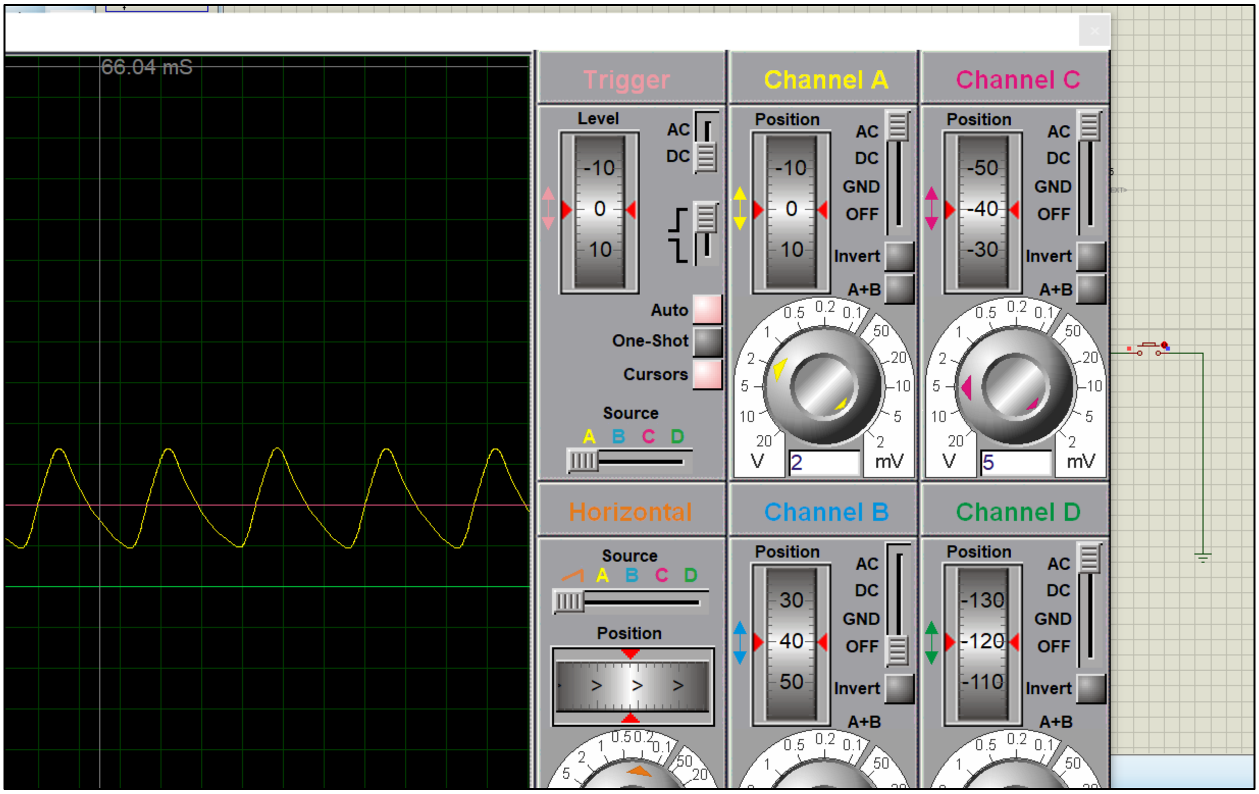
main.c

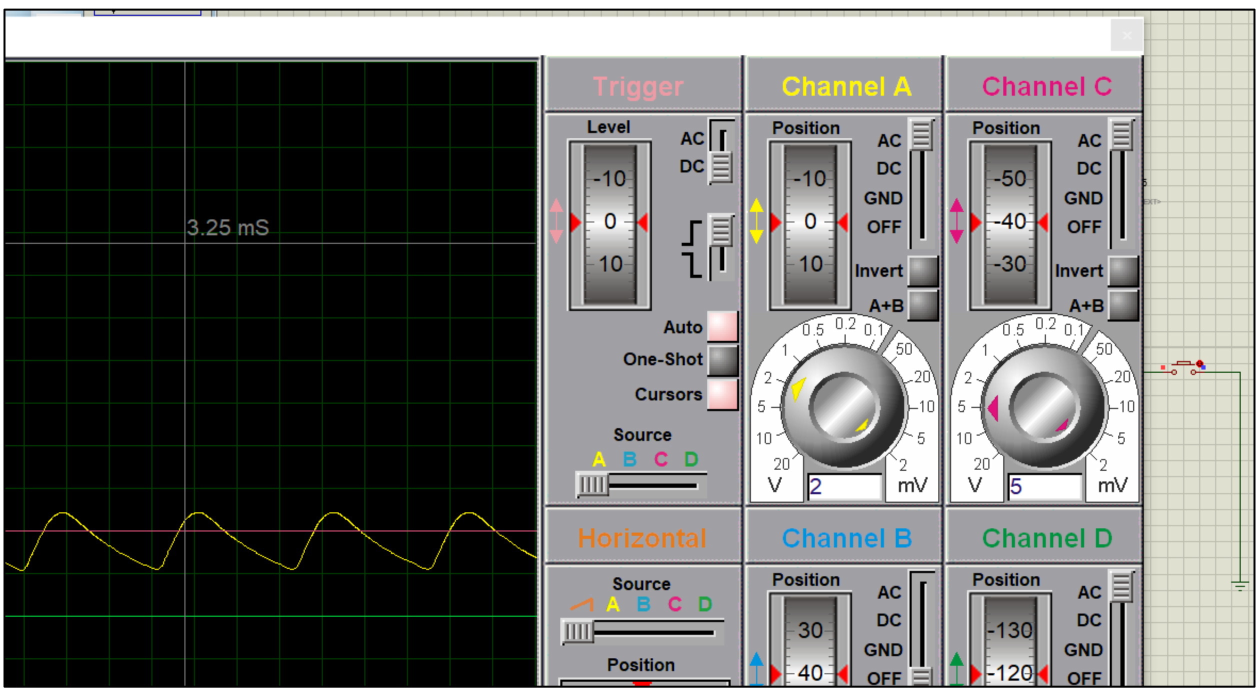
```c
void squarewave(){

	P1=0x00;
	delay();
	P1=0xFF;
	delay();
}

void sawtoothwave(){
	char sawtooth_val[] = {255, 233, 210, 180, 132, 100, 76, 54, 20, 13, 0 };

	for(i = 0; i<11; i++)
	    P1 = sawtooth_val[i];
}

void triangularwave(){

	char triangular_val[] = {0, 4, 8, 16, 32, 64, 128, 255, 128, 64, 32, 16, 8, 4, 2, 0};

	for(i = 0; i<16; i++)
		P1 = triangular_val[i];
}


void delay(){
      for(i = 0; i<3000; i++);
}
```

Proteus Simulation:






66.04 mS
Trigger
Level
AC
DC
Auto
One-Shot
Cursors
Source
A B C D
Channel A
Position
AC
DC
GND
OFF
Invert
A+B
V
2
mV
Channel C
Position
AC
DC
GND
OFF
Invert
A+B
V
5
mV
Horizontal
Source
A B C D
Position
Channel B
Position
AC
DC
GND
OFF
Invert
A+B
Channel D
Position
AC
DC
GND
OFF
Invert
A+B


3.25 mS
Trigger
Level
AC
DC
Auto
One-Shot
Cursors
Source
A B C D
Channel A
Position
AC
DC
GND
OFF
Invert
A+B
V
2
mV
Channel C
Position
AC
DC
GND
OFF
Invert
A+B
V
5
mV
Horizontal
Source
A B C D
Position
Channel B
Position
AC
DC
GND
OFF
Channel D
Position
AC
DC
GND
OFF